

MGT 3213 - Management Information Systems

S # 5 - Information Systems Planning

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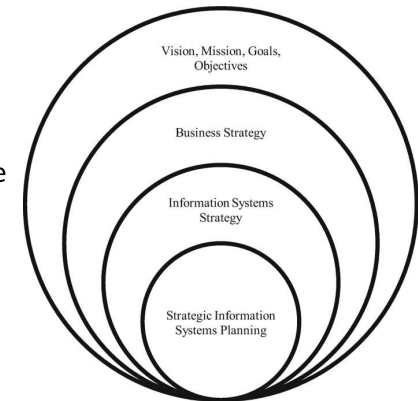
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What is Information Systems Planning?

- Information Systems Planning (ISP) is a strategic, structured process that aligns an organization's IT infrastructure and applications with its overall business goals, goals, and strategies.
- It involves developing a roadmap to optimize information, technology, and resources for competitive advantage, efficiency, and improved decision-making.
- Key components include assessing the current and future environment, defining technical architecture, and prioritizing implementation.



Information Systems Planning in an organization

- ISP requires support of top management and its commitment to reach desired objectives.
- Techniques used to capture information system needs, include
 - Interviewing managers and executives
 - Reviewing corporate documents
 - Analysing competitors, markets and products
- Three key activities:
 - Describe the Current Situation
 - Describe the Target (or Future) Situation
 - Develop a Transition Plan and Strategy

Information Systems Planning in an organization

- Information Systems Planning
 1. Describing the Current Situation

There are two approaches to describe current situation needs

 - Top-down Planning
 - Generic methodology that attempts to gain a broad understanding of the information system needs of the entire organization
 - Bottom-up Planning
 - Generic methodology that identifies and defines IS development projects based upon solving operational business problems or taking advantage of some business opportunities

Information Systems Planning in an organization

1. Describing the Current Situation (Continued)

Include :

- Planning team is chartered to model existing situation.
- Identification of Organizational:
 - Locations
 - Units
 - Functions
 - Processes
 - Data
 - Information Systems

Information collected about current situation

- **Locations** where the company operates
- **Business unit** that operate within a company
- **Functions** = cross organisational collection of activities used to perform day-to-day business operation.
- **Processes** of a company
- **Data** underlying processes
- **Information system** represent automated and non automated systems used to support business processes.

Information Systems Planning in an organization

2. Describing the Target Situation

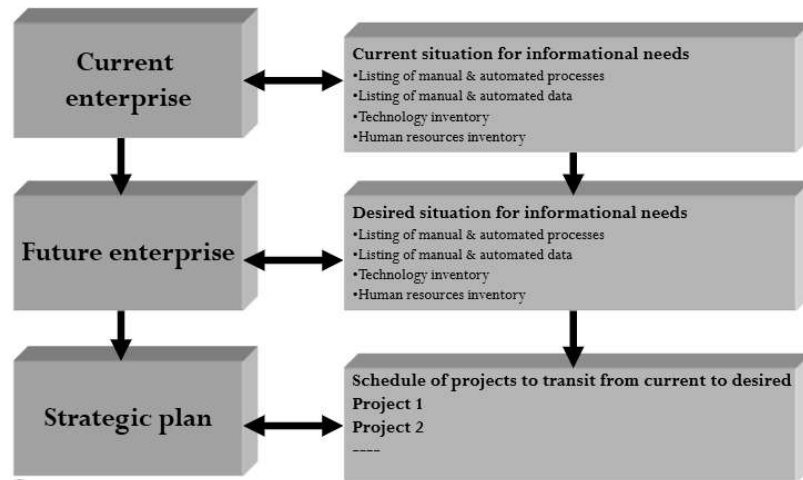
- Define situation and update list of organizational locations, functions, etc. to reflect desired locations, functions, etc.
- Planners focus on differences between current lists and future lists and

Information Systems Planning in an organization

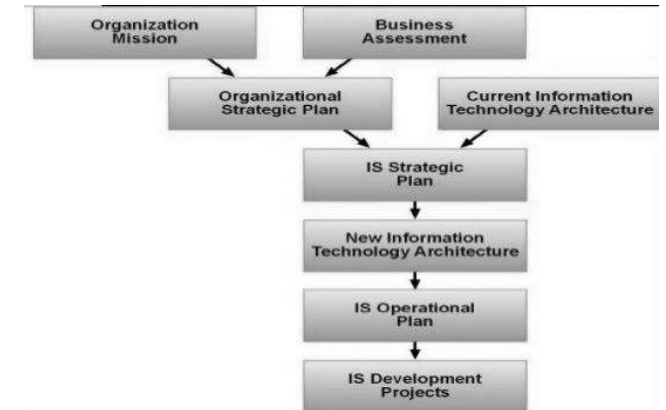
3-Developing a transition strategy and plans

- Plans reflect both short-term and long-term organisational development needs
- Plans include organisational mission, information inventory, mission and objectives of IS, constraints, long-term plans and short-term plans
- Selected projects are those derived from ISP and that fulfil the gap between current and desired situation

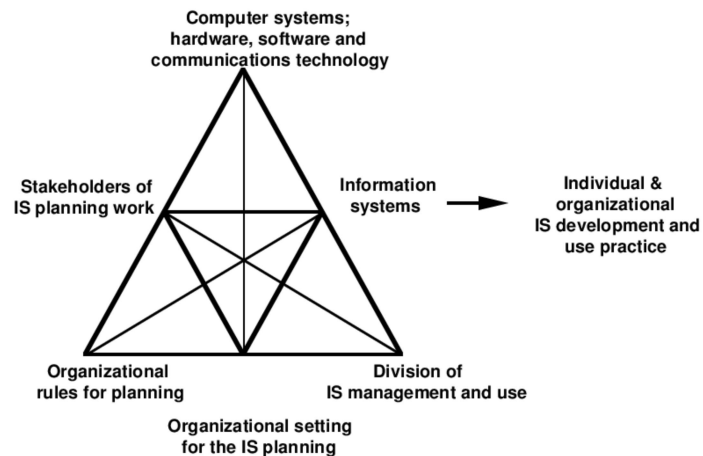
ISP and Corporate Strategy Planning



IS Planning Process



The structure of strategic information systems planning in organizations



Line Organisational Structure

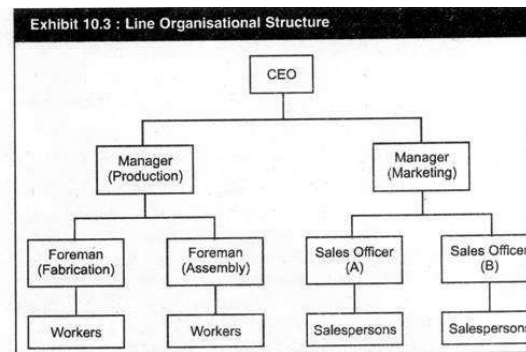
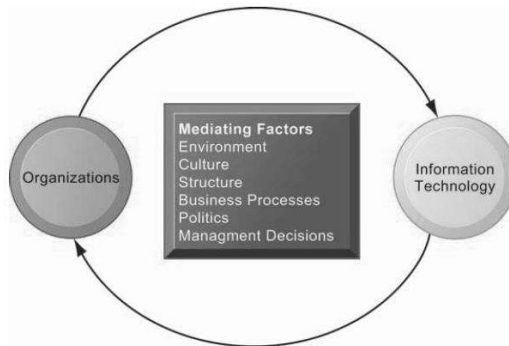
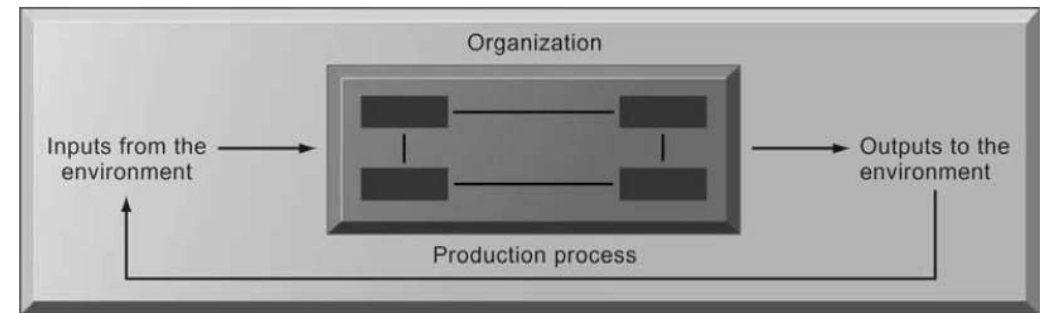


FIGURE 3.1 THE TWO-WAY RELATIONSHIP BETWEEN ORGANIZATIONS AND INFORMATION TECHNOLOGY



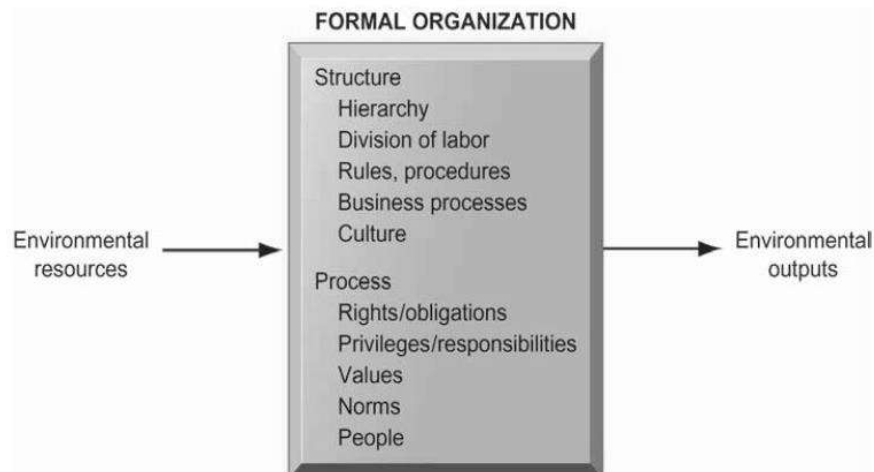
This complex two-way relationship is mediated by many factors, not the least of which are the decisions made—or not made—by managers. Other factors mediating the relationship include the organizational culture, structure, politics, business processes, and environment.

FIGURE 3.2 THE TECHNICAL MICROECONOMIC DEFINITION OF THE ORGANIZATION



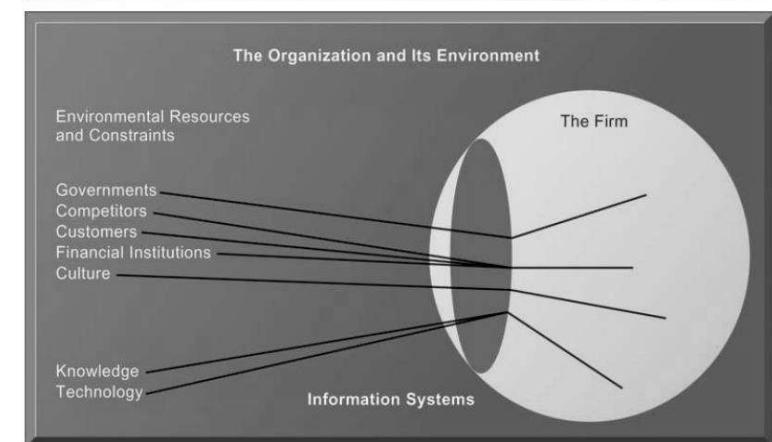
In the microeconomic definition of organizations, capital and labor (the primary production factors provided by the environment) are transformed by the firm through the production process into products and services (outputs to the environment). The products and services are consumed by the environment, which supplies additional capital and labor as inputs in the feedback loop.

FIGURE 3.3 THE BEHAVIORAL VIEW OF ORGANIZATIONS



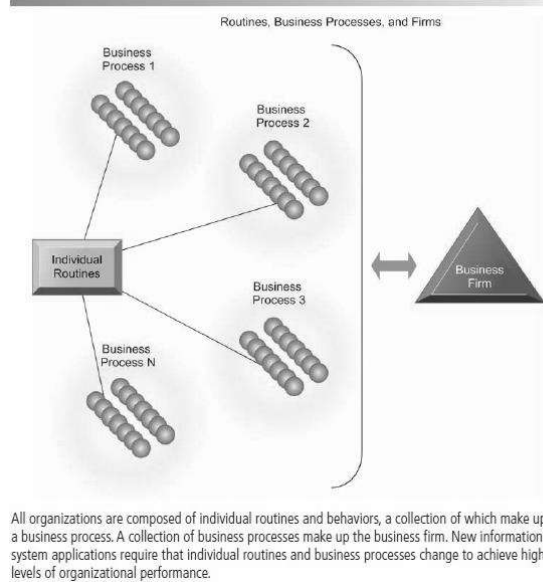
The behavioral view of organizations emphasizes group relationships, values, and structures.

FIGURE 3.5 ENVIRONMENTS AND ORGANIZATIONS HAVE A RECIPROCAL RELATIONSHIP

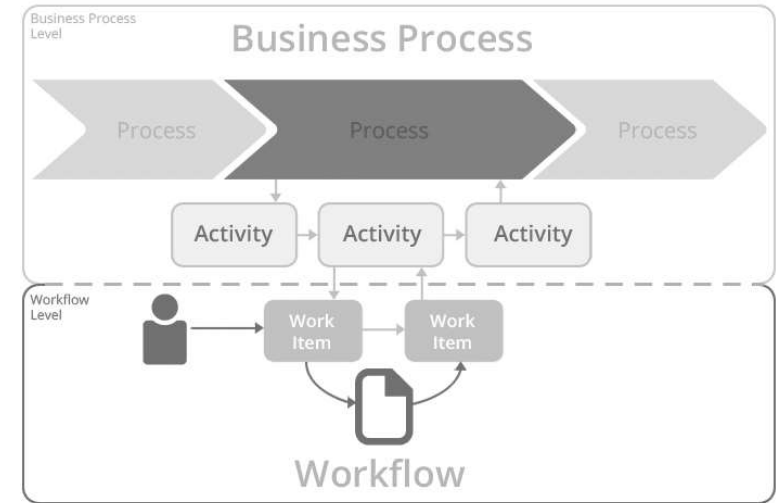


Environments shape what organizations can do, but organizations can influence their environments and decide to change environments altogether. Information technology plays a critical role in helping organizations perceive environmental change and in helping organizations act on their environment.

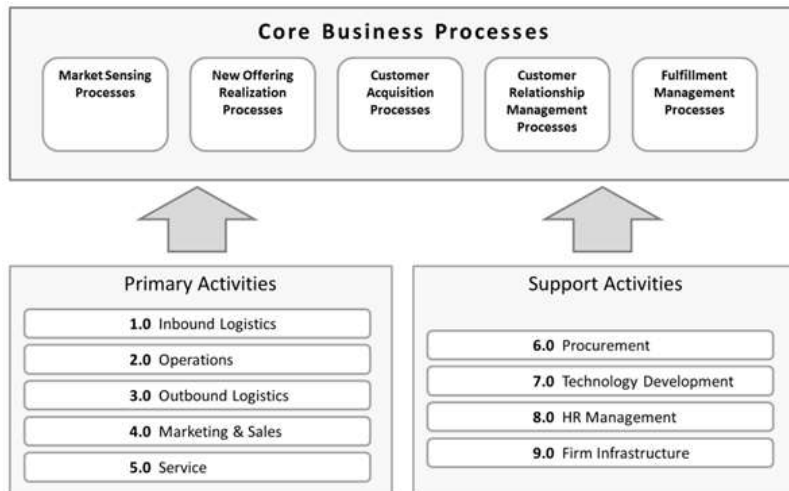
FIGURE 3.4 ROUTINES, BUSINESS PROCESSES, AND FIRMS



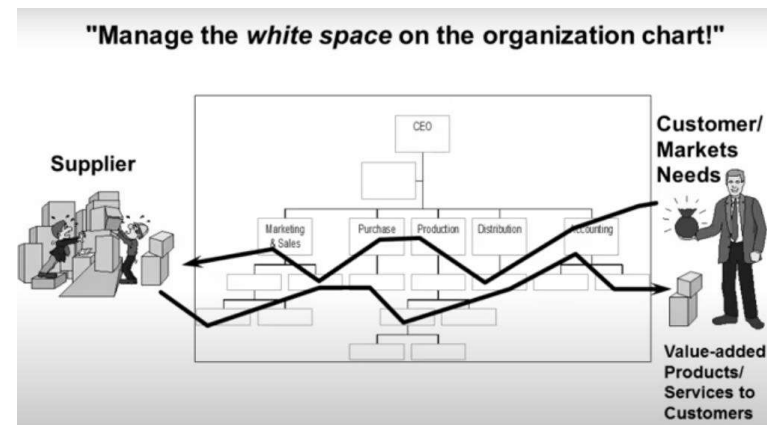
Business Process and Workflow



Core Business Process



Process are often cross functional areas



- We cannot improve or measure the performance of a hierarchical structure.
- But, we can increase **output quality** and **customer satisfaction**, as well as reduce the **cost** and **cycle time** of a process to improve it

How to do Business Process Analysis

Thank you

Business Process Analysis is done through the following seven steps.

Step 1 - Define the Process

Step 2 - Uncover Opportunities

Step 3 - Measure for Success

Step 4 - Analyze the Process

Step 5 - Take Effective Action

Step 6 - Establish a State of Control

Step 7 - Monitor for Effectiveness